Heterogeneous Micro-Fusion: Reconstructing Frozen Pretrained Models at the Feed-Forward Sublayer

Anonymous Author(s)
Affiliation
Address
email

DEPRECATED — MERGED. This draft has been merged and restructured into `reconstruct.tex` (English) and `reconstruct_zh.tex` (Chinese), in this directory. It is kept for archival only: its numbers, section structure, and claims are superseded by the merged manuscript. Do not cite this file.

Abstract

Combining heterogeneous pretrained models—bidirectional encoders, causal decoders, and vision transformers that differ in architecture, hidden width, and modality—usually requires joint retraining, architecture surgery, or lossy output-level ensembling. We study a bottom-level alternative that operates where such models are most exchangeable: the feed-forward sublayer. Our method, **heterogeneous micro-fusion**, routes each token to one of M frozen architectural experts through a shared sparse Top-K router trained with a straight-through estimator, injects INT4 pseudo-quantization noise into expert outputs, and fuses experts residually with per-expert gates, while every base model stays frozen. A cross-architecture semantic aligner maps mismatched hidden spaces into one routing space; across four aligner designs we measure a strict performance ladder (linear $<$ mean $<$ attention $<$ mixture) that saturates at the mixture aligner, and nesting it brings no further gain; at CPU scale we report this ceiling as a hypothesis rather than a law. Because fused inference still requires every base to stay alive, we add a second reconstruction step, **chimera distillation**, which collapses the ensemble into one independent base: a real TinyBERT student with 0.60% trainable parameters surpasses its multi-base teacher by 50.1% in reconstruction MSE under a mixed objective that includes a direct target signal, and a multimodal BERT+ViT student surpasses it by 50.0%; the v32 ablation shows that pure distillation—with the direct target signal removed—does not outperform the teacher, so the entire measured margin is attributable to the leaked target term rather than to concept-direction transfer. We report the negative results that bound the method—component composition, nested aligners, multi-teacher distillation, and toy multimodal benchmarks—from hundreds of seeded end-to-end runs across 25 engineering iterations.

1 Introduction

Practitioners rarely have the luxury of one perfect pretrained model. A working pipeline may need a bidirectional encoder for text understanding, a causal decoder for generation, and a vision transformer for perception. Three ways to combine such models dominate current practice, and each is unsatisfying. Joint multi-task retraining is expensive and re-opens the pretraining budget. Output-level ensembling keeps every model alive at inference and blends predictions that were never trained to complement each other. Weight merging [Ilharco et al., 2023, Yadav et al., 2023] requires the

Submitted to 39th Conference on Neural Information Processing Systems (NeurIPS 2025). Do not distribute.

merged models to share architecture and initialization, which rules out heterogeneous bases outright.

This paper studies a fourth option, one level lower. Within a transformer block, the feed-forward (FFN) sublayer is the most modular and the most replaceable component: it acts position-wise, it carries a large share of the parameters, and its role as a key–value memory [Geva et al., 2021] suggests that *which* memory bank is consulted can be decided per token. We therefore reconstruct the FFN sublayer itself as a **heterogeneous micro-fusion layer**: a shared, sparse Top-K router selects among M frozen architectural experts; rectangular projections or learned semantic aligners map mismatched hidden spaces ($D_m \in \{192, 312, 768\}$) into one shared routing space; per-expert gates, channel-wise alignment, and INT4 pseudo-quantization noise complete the layer. No base model is modified, and only alignment, routing, and gate parameters are trained.

Position in the heterogeneity spectrum. Recent work studies pretrained-model collaboration along three axes: architecture (same vs different), modality (text vs vision vs multimodal), and routing granularity (token-level vs query-level). In the architecture-homogeneous limit, sparse upcycling and its industrial descendants—Mixtral [Jiang et al., 2024], Symphony-MoE [Wang et al., 2025], Branch-Train-MiX [Sukhbaatar et al., 2024], Soft MoE [Puigcerver et al., 2024], and shared-and-fine-grained-expert designs [Dai et al., 2024]—scale to industrial sizes by exploiting that every expert shares the same dimension, tokenizer, and hidden-space geometry. The architecture-heterogeneous limit is sparsely populated: the closest published antecedent to our fusion layer is ScaleKD's cross-attention projector with dual-view (CLS + spatial) mimicking [Wang et al., 2024b]. For orchestrated heterogeneous LLM inference in cross-border e-commerce relevance, query-level coarse-grained routing over heterogeneous LLM experts with anisotropy-preserving concatenation fusion [Liu et al., 2026] independently argues that weighted averaging collapses distinct embedding manifolds—an empirical claim our four-mode comparison corroborates. Expert-aware post-training quantization [Du et al., 2026], which validates the cross-expert channel-similarity assumption, and multi-teacher LLM distillation [Tian et al., 2025], which recovers knowledge diversity through rationale-augmented objectives, address adjacent compression and training problems; the non-destructive composition philosophy we adopt for cross-architecture fusion is shared with adapter composition [Pfeiffer et al., 2021, Huang et al., 2024]. We sit at the intersection of these lines: every axis is at its hardest, the deployment scale is deliberately CPU-sandboxed rather than industrial, and the heterogeneity spectrum makes a structural prediction—token-level routing has no analogue across architectures, so the coarse-grained query-level gate that emerges is the only routing granularity that composes naturally with frozen, mismatched bases.

Fusion solved at the sublayer level has a deployment cost: the fused system still needs every base alive at inference. We close the loop with a second reconstruction step that we call **chimera distillation**. A single-base student—a real frozen TinyBERT [Jiao et al., 2020] with a low-rank adapter [Hu et al., 2022]—is trained to reproduce the fused teacher's output from its *own* modality's input alone. The distilled student surpasses its multi-base teacher by 50.1% in reconstruction error while keeping 99.4% of its parameters frozen, and a multimodal BERT+ViT student surpasses the same teacher by 50.0%; the v32 ablation (§5.4) shows that this margin is entirely attributable to the 0.3 direct-target term in the distillation loss, not to internalization of the teacher's concept directions. The mechanism evidence therefore lies in the cross-student comparison under the identical loss, not in the absolute student–teacher margin.[1]

The study is deliberately systematic rather than heroic. All experiments run in a CPU-only sandbox at small width ($D_{\text{shared}}=256$, sequence length 16), with five seeds per configuration and every claim backed by an executable script and unit tests. Within that scope we map three territories:

1. **Sublayer-level fusion across truly heterogeneous bases.** With per-modality native encoding, rectangular projections, and a shared sparse router, fusion beats the best single expert by 20.3% and equal-weight averaging by 17.3%. Making the attention sublayers routable as well raises the cross-architecture margin over averaging to 84.8%, and a path decomposition shows the gain concentrates in the attention path: cross-architecture differences surface mainly in attention patterns, whose behavior is far more architecture-specific than FFN activations.

[1]The name borrows the chimera of myth—one body assembled from parts of different animals—for an ensemble assembled from heterogeneous bases: the student inherits the assembled behavior in a single body.

2. **An alignment ladder with a measured ceiling.** Replacing the frozen linear projection with a cross-base mean, then shared-K/V attention, then a mixture-of-experts aligner yields a strict ladder ($1.14 \to 0.48 \to 0.34 \to 0.26$ MSE). Nesting the mixture aligner does not improve on it: the ladder saturates at one level of cross-expert attention.

3. **Independent-model reconstruction by distillation.** A distilled student with 0.60% trainable parameters beats its teacher by 50.1% under a mixed objective that includes a direct target signal; a corrected evaluation on held-out real Wikipedia text keeps the route viable (held-out MSE 0.157); a multimodal student beats it by 50.0%. The v32 ablation shows that pure distillation does not outperform the teacher, so the entire margin is attributable to the 0.3 direct-target term in the loss. Multi-teacher distillation, by contrast, regresses across every configuration we tried.

Just as informative are the boundaries, five distinct negative results in total. (i) Composing a repaired router with a stronger aligner adds nothing once the aligner already contains the repair—$+71.0\%$ vs. $+72.6\%$ gain over their common baseline, less than two percentage points apart, but consistently in the aligner's favor across all 5 seeds. (ii) Harder toy VQA benchmarks do not rescue multimodal evaluation—all models drop to random accuracy. (iii) Multi-teacher distillation regresses in every configuration tried (§5.4). (iv) Nesting the mixture aligner with itself regresses by 4.7% MSE. (v) The v32 ablation shows that pure distillation does not outperform the teacher—so the entire measured student-vs-teacher margin is attributable to the 0.3 direct-target term in the distillation loss rather than to genuine concept-direction transfer. The recurring lesson is that adding machinery is not the same as adding capability.

2 Related Work

Mixture-of-experts routing. Conditional computation routes each input to a subset of experts [Jacobs et al., 1991, Shazeer et al., 2017], and modern sparse MoE stacks scale this to trillion-parameter regimes [Fedus et al., 2022, Zoph et al., 2022]. Industry-scale deployments such as Mixtral 8x7B [Jiang et al., 2024] deliver 47B total / 13B active parameters per token via top-2 token-level routing over eight FFN copies, with measured *temporal locality* (62% of adjacent tokens routed to the same expert) and clear domain specialization across mathematics, code, and multilingual data. Token-level soft routing addresses MoE's instabilities and load imbalance via slot-based differentiable mixtures [Puigcerver et al., 2024]. Fine-grained expert segmentation together with shared-expert isolation [Dai et al., 2024]—and the more general recipe of *branch-and-train then mix* as experts, finetuned into a routed MoE [Sukhbaatar et al., 2024]—further refines specialization. In all of these systems, however, the experts are *homogeneous*: they share architecture, dimension, and training. Sparse upcycling [Komatsuzaki et al., 2023] converts a dense checkpoint into an MoE by cloning and perturbing its FFNs—again producing homogeneous experts. Our experts are the opposite: they are entire pretrained models with different architectures, tokenizers, and modalities, kept frozen, and the routing problem becomes one of aligning incompatible representation spaces rather than balancing identical ones.

Heterogeneous foundation-model collaboration. A growing line of work studies collaboration between pretrained models that do *not* share architecture. Within the homogeneous limit, Symphony-MoE [Wang et al., 2025] upcycles several like-architected but differently-pretrained LLMs (Qwen2.5-Coder + Qwen2) by harmonizing shared backbones via SLERP and re-permuting FFN neurons through a linear assignment problem; the harder cross-architecture case is studied under cross-architecture distillation [Wang et al., 2024b], whose cross-attention projector plus dual-view (CLS + spatial) mimicking is the closest published antecedent to our fusion layer. For orchestrated heterogeneous LLM inference in cross-border e-commerce relevance, orchestrating heterogeneous LLM experts with query-level routing and anisotropy-preserving concatenation fusion [Liu et al., 2026] independently shows that *concatenating* expert embeddings beats weighted averaging for models with distinct embedding manifolds—a geometric argument our four-mode comparison (§3) corroborates. Expert-aware post-training quantization [Du et al., 2026] validates the *cross-expert channel similarity* assumption that underpins our v22 ChannelNorm. Together these works delineate the heterogeneity spectrum within which we sit: same architecture, same modality, same manifold is the easy limit; our setting is simultaneously different on all three axes.

Model merging and upcycling. Task arithmetic [Ilharco et al., 2023], TIES-merging [Yadav et al., 2023], and model soups [Wortsman et al., 2022] combine *weights* of models that share a backbone. Heterogeneous bases cannot be merged this way; our fusion layer is the corresponding mechanism for the heterogeneous case, operating through activations instead of parameter addition.

Parameter-efficient adaptation. Adapters [Houlsby et al., 2019] and LoRA [Hu et al., 2022] freeze a base and train small insertions. We use the same philosophy, but the frozen part is a *pool* of heterogeneous bases, and the trained part is the inter-base machinery (projections, aligners, router, gates). AdapterFusion [Pfeiffer et al., 2021] composes per-task adapters non-destructively through a learned Q/K/V attention combiner, and Huang et al. [2024] generalize this idea to dynamic linear combination of per-task LoRA modules optimized via gradient-free black-box search on a few-shot target task—both demonstrate the *non-destructive composition* principle we independently adopt for cross-architecture fusion. Our aligner ladder can be read as an activation-level analogue, and our multi-teacher negative result (§5.4) echoes Pfeiffer et al.'s finding that naive source combination is not free. Pfeiffer et al. (2021) further observe that AdapterFusion outperforms full fine-tuning in low-data regimes; our four-mode comparison is consistent with the non-destructive composition principle they articulate, but is run at a single training scale and does not by itself demonstrate data-scale dependence.

Knowledge distillation. Distillation transfers a teacher's behavior to a smaller student [Hinton et al., 2015, Sanh et al., 2019, Gou et al., 2021]. Classic pipelines distill one teacher into a student of the same modality, and multi-teacher variants are known [You et al., 2017]; recent multi-teacher LLM distillation uses rationale-augmented objectives and teacher-forcing to recover knowledge diversity [Tian et al., 2025]. Chimera distillation differs in both directions: the teacher is a multi-base *ensemble*—echoing the collective-intelligence framing of mixture-of-agents [Wang et al., 2024a], though our collaboration operates at the representation level through learned routing and gating rather than at the output level—and the student sees only one base's input yet must reproduce the ensemble's fused output, forcing it to internalize information from modalities it never observes. Quantization-aware training injects fake-quantization noise during training [Dettmers et al., 2022]; we use INT4 fake quantization on expert outputs as regularization and hardware-realism pressure, not as a deployment compression claim.

3 Method

3.1 Problem setup

Let $\mathcal{B} = \{B_1, \ldots, B_M\}$ be M frozen pretrained bases with hidden sizes D_m and, in general, different architectures: a bidirectional encoder (TinyBERT, D=312) [Jiao et al., 2020, Devlin et al., 2019], a causal decoder (TinyLlama-110M, D=768), and a vision transformer (ViT-tiny, D=192) [Dosovitskiy et al., 2021]. Each base consumes its own tokenizer or processor and produces hidden states $h_m \in \mathbb{R}^{S \times D_m}$ natively; we never force a shared encoder. The goal is a fusion layer operating in a shared space of width $D_{\text{shared}} = 256$ such that (i) every base remains frozen, (ii) only alignment, routing, and gate parameters are trained, and (iii) the layer degrades gracefully—disabling any single expert leaves a working system.

3.2 The heterogeneous fusion layer

Contributions of the method, ranked by support strength. The architecture is shaped by three operationally supported contributions and one hypothesis-tier observation; we cite them as C1–C4 throughout, with C2 carrying the strongest falsifiable finding and C3 explicitly framed as a literature-convergent hypothesis rather than a method-level contribution. (**C2**) *Aligner ladder ceiling* (**main finding**). §3.4 reports a strict ladder of four alignment designs—linear $<$ mean $<$ cross-architecture attention $<$ mixture-of-experts—evaluated under matched conditions on three frozen heterogeneous bases. The ladder *saturates* at the mixture aligner, and nesting the mixture—a design that strictly increases parameter count—regresses by 4.7% MSE. We treat this ceiling as the paper's empirical contribution: it is a measured, falsifiable boundary on heterogeneous-base composition that the architecture-level discussion in §6 evaluates against alternative routing schemes. (**C1**) *Gate-style sparse top-K routing as the enabling mechanism.* The single sparse router and per-expert gates described in §3.2–§3.3 are not a learned combination matrix but a hard selection plus a per-expert scale; this is what makes C2's ladder trainable on CPU with frozen bases—a non-trivial constraint given

that cross-architecture routing has historically required either full backpropagation through heterogeneous parameter sets or output-level ensembling. The mechanism is shared with the sparse-MoE literature [**??**Jacobs et al., 1991] but here serves cross-architecture rather than within-architecture expert selection. (**C4**) *Dual-path sublayer routing with scale-heterogeneous bases* (**design extension**). The C1 router is applied to both attention and FFN sublayers (§3.5); the two pathways may select experts from *different* bases with different widths, modalities, and inductive biases—this is the scale-heterogeneous half of C4. The attention–FFN decomposition experiment in §5.2 shows that cross-architecture complementarity concentrates in the attention path. The public evidence for scale-heterogeneous selection rests on dual-base configurations; the larger-scale half of C4 uses proprietary auxiliary models and is held author-only (see the project README note on `research/scale_heterogeneous/`). (**C3**) *Data-scale dependence is a literature-convergent hypothesis* (**observation tier**). Four independent lines of recent work on routing and composition report that the optimal composition of heterogeneous experts shifts with training-data scale (§6). Our four-mode ladder is run at a single training scale and is consistent with, but does not by itself establish, that regime boundary; we report C3 as an observation in support of a hypothesis rather than as a method contribution. An in-paper data-scale sweep is left to future work. Figure 3 summarizes the contribution structure.

Figure 2 shows the layer. Given the residual-stream input $x \in \mathbb{R}^{B\times S\times D_{\text{shared}}}$, each expert m computes

$$h_m = \mathrm{sg}(x) \odot \gamma_m + \beta_m, \qquad \tilde{F}_m = Q_{\text{INT4}}\big(F_m(h_m)\big), \tag{1}$$

$$F_m(h) = \big(\mathrm{SiLU}(hW_m^{\text{gate}}) \odot (hW_m^{\text{up}})\big)\, W_m^{\text{down}}, \tag{2}$$

where Q_{INT4} is group-wise (group size 128) 4-bit fake quantization with a straight-through estimator in the backward pass. Three deliberate choices shape the layer:

Gradient truncation. The expert input is detached (sg), so the frozen bases' intermediate activations never enter the autograd graph. Activation memory drops from $O(MLD_{\text{ff}})$ to $O(LD_{\text{shared}})$ plus adapter parameters, which is what makes M-base fusion trainable on CPU at all. The cost is that experts cannot shape upstream representations through gradients; the router and gates carry all adaptation.

Sparse routing with straight-through gradients. The router computes logits $z = xW_r^\top$ from the *clean* pre-noise input, $\alpha = \mathrm{softmax}(z)$, and the hard selection $\hat{\alpha} = \mathrm{TopK}(\alpha, K)$ (K=1 unless stated) is applied in the forward pass while the backward pass follows the softmax weights, in the spirit of straight-through estimators [Bengio et al., 2013]. Routing on the clean input keeps router decisions from chasing the quantization noise it would otherwise induce.

Residual fusion with positive gates. The layer output is

$$y = x + \sum_{m=1}^{M} \hat{\alpha}_m\, \alpha_m\, \tilde{F}_m(h_m), \tag{3}$$

with $\alpha_m \sim \mathcal{U}(0.01, 0.1)$ at initialization. The positive, non-trivial initialization avoids the near-zero-gate dead lock: gates initialized at 10^{-4} scale receive gradients too small to escape, which we observed as flat training curves before adopting the uniform initialization. These three choices together operationalize Innovation C1: a gate-style sparse top-K routing that is trainable on CPU with frozen bases.

3.3 Two-phase training protocol

Routing is fragile early in training—random router weights produce noise selections that poison the expert gates. Training therefore runs in two phases. **Phase 1** (first 10% of steps): routing is forced uniform ($\hat{\alpha}_m = 1/M$, detached) and the router is frozen, so only alignment and gate parameters learn. **Phase 2** (remaining 90%): the router unlocks and Top-K selection activates, with the loss

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_1 \mathcal{L}_{\text{balance}} + \lambda_2 \mathcal{L}_z, \tag{4}$$

where $\mathcal{L}_{\text{balance}}$ encourages even expert load and $\mathcal{L}_z$ regularizes router logits. Optimizer groups isolate learning rates: 10^{-4} for the router, 10^{-2} for alignment parameters, and 10^{-3} for the output gates α_m. The router must move slowly because its selections are discrete; the alignment layers can move quickly because their gradients are dense. This group isolation is the operational expression of Innovation C1's constraint that *only* alignment, routing, and gate parameters are trained.

3.4 The alignment ladder

The shared space is where heterogeneous bases meet, and how it is constructed turned out to matter more than anything else in the stack. We evaluate four designs of increasing sophistication (Figure 4):

Linear projection (baseline). Each base's hidden state passes through a rectangular projection $P_m \in \mathbb{R}^{D_{\text{shared}} \times D_m}$, orthogonal-initialized and frozen. Bases are projected independently and know nothing about each other.

Mean aligner. Learned per-base down-projections $W_m \in \mathbb{R}^{d \times D_m}$ followed by a cross-base mean of the projected representations.

Cross-architecture attention aligner. Each base is projected to the shared width; the cross-base mean provides shared keys and values; each base queries the shared context with its own query projection—the standard query/key/value separation of cross-attention [Vaswani et al., 2017]—and adds the attended concept vector residually. The per-base results are averaged. Every base therefore reads a concept summary written by *all* bases.

Mixture aligner. Each base contributes $n{=}4$ small shared experts (two-layer MLPs) to a pooled bank of $M \times n$ experts. Expert outputs serve as queries against a shared key/value context derived from their mean, and the attended outputs are aggregated. The aligner itself becomes a tiny mixture-of-experts over the heterogeneous sources.

Two empirical facts motivate the ladder. First, *native encoding is not optional*: in an early control where all three modalities were tokenized and encoded by the same BERT, the fusion gain over a single expert collapsed from $+20.3\%$ to $+4.0\%$. Uniform encoding erases exactly the architectural differences the router is supposed to exploit. Second, the ladder is *strict but saturating*: each mode improves on the previous one, and the improvement stops at the mixture aligner (§5.3)—a ceiling we hold as a scale-bounded hypothesis, not a law.

Four modes, in one sentence (Innovation C2). We label the four modes A / B / C / D (linear / mean / cross-attention / mixture) and report their MSE under matched conditions in §5.3; the strict-but-saturating property is what licenses us to call this a *systematic comparison* rather than a single best design. We treat the data-scale dependence of the winning mode as a literature-convergent hypothesis (Innovation C3); the held-out real-corpus evaluation in §5.3 varies only the evaluation distribution, not the training-data scale, and an in-paper sweep across training scales is left to future work.

3.5 Dual-path routing: making attention routable

The FFN-only layer routes among memory banks but always consumes the same attention output. The dual-path extension makes the attention sublayers routable too: each token independently selects one *attention expert* and one *FFN expert*, from two pools that may come from different bases. Attention experts expose the native Q/K/V/O projections of the base's first block; the selected expert's attention is recomputed in its native space and projected back through P_m. Per-expert optimizer groups (11+ groups in one AdamW) implement the learning-rate isolation of §3.3 per expert. The dual-path design asks whether cross-architecture complementarity lives in the FFN weights, in the attention patterns, or in both—and the path decomposition in §5.2 gives a sharp answer.

Dual-path sublayer routing (Innovation C4). Read in mechanism terms, the attention pathway runs composed attention on the chosen expert and the FFN pathway runs the gate-style sparse top-K router's selection; the two need not pick the same expert, and the *scale-heterogeneous* half of Innovation C4 comes from letting them come from different bases with different widths, modalities, and inductive biases. Both pathways are gated, residual, and trained with the same sparse router and the same straight-through gradient on the clean input; only the experts themselves are frozen. The path-decomposition result in §5.2 is the empirical justification for the design; the framing makes the architecture-level commitment explicit: *two* sparse routes, not one.

4 Reconstructing an independent model

Fusion at the sublayer level keeps every base on the inference path. For deployment we want the opposite: one model, one modality, no ensemble. The chimera distillation step compresses the ensemble into a single-base student that never sees the other bases' inputs. The distillation protocol is also where the data-scale hypothesis (Innovation C3) is most directly testable: the v25–v29 protocol varies training-corpus size and could in principle demonstrate the regime boundary; the in-paper sweep is left to future work.

Student architecture. The unimodal student is a real frozen TinyBERT [Jiao et al., 2020] with a rank-8 LoRA adapter (α=16) applied to its hidden states, a linear head to D_{shared}, and a LayerNorm. Only $85{,}632$ of $\sim$14M parameters (0.60%) are trainable. Unlike the simplified MLP student we first tried, a pretrained student brings prior linguistic structure to the reconstruction problem.

Distillation objective. The teacher is the fused multi-base system with the mixture aligner. Given a batch from the *student's modality only*, the teacher runs its full multi-base forward pass and the student must match it:

$$\mathcal{L}_{\text{distill}} = \alpha_{\text{h}}\,\text{MSE}\big(y_s,\,\text{sg}(y_t)\big) + \beta_{\text{h}}\,\text{MSE}\big(y_s,\,y_{\text{target}}\big), \tag{5}$$

with $\alpha_{\text{h}} = 0.7$ and $\beta_{\text{h}} = 0.3$. The first term distills the teacher's hidden-state concept directions; the second adds a direct target signal. This is not a temperature-scaled KL divergence but a weighted combination of two MSE terms. We quantify the contribution of the direct-target term in §5.4 via the **pure** condition (`run_v32`), which sets $\beta_{\text{h}} = 0$ and isolates pure distillation from target leakage.

Multimodal student. The multimodal student pairs TinyBERT (text) with ViT-tiny (image), each with its own LoRA, plus a learned modality router that decides which base processes which input. It is trained with the same objective against the same single-teacher ensemble.

Evaluation discipline. Our first real-student experiments were evaluated on training-distribution inputs with easily fitted targets and produced near-zero MSE—an overfitting artifact masquerading as success. All later student evaluations use held-out inputs from a real mini-corpus (16 Wikipedia-style training sentences, 6 held out), forcing the student to reproduce the teacher's concept directions under a token distribution it never trained on. We flag this correction explicitly because the uncorrected numbers are flattering to the point of meaninglessness.

5 Experiments

5.1 Setup and honest scope

All experiments run on CPU only (Linux, PyTorch 2.x, no GPU acceleration). Widths are small (D_{shared}=256, S=16), targets are synthetic class-direction constructions (a block of the shared space is set to 1.0 for the target class), and every configuration is averaged over 5 seeds unless stated. The full stack passes 430 unit and integration tests; each number below is produced by a named, committed script (reproduction commands in §8). These are *mechanism-level* results: they establish that the reconstruction mechanisms behave as designed under controlled conditions, not that the method survives 0.5B-parameter scale or perplexity-level evaluation on real tasks. We consider the scale gap the single largest open risk and discuss it in §7.

5.2 Sublayer fusion gains

Table 1 summarizes the fusion results. With FFN-only routing (v6), the fused layer beats the best single expert by 20.3% and equal-weight averaging by 17.3%, with every seed passing—the first demonstration that genuinely different pretrained architectures can share one sparse router at the sublayer level. Dual-path routing (v7-A) raises the margin over averaging to 84.8%.

The path decomposition is the interesting part. In the cross-architecture setting, the attention path alone reaches MSE 1.16—already close to the fused 1.06—while the FFN path alone (7.10) is indistinguishable from averaging (7.06). Cross-architecture complementarity therefore surfaces almost entirely in *attention patterns*: once each token can consult the architecture whose attention mode fits it, the FFN experts' remaining contribution is the residual refinement. We state this narrowly—attention varies more across architectures than FFN activations do—and it does not contradict the

Table 1: Sublayer-level fusion of heterogeneous bases (5-seed means, fuse MSE; lower is better). v6: FFN-only routing over three architectures. v7-A/v7-B: dual-path routing (one attention + one FFN expert per token), cross-architecture and same-architecture respectively. Gain is computed per seed then averaged.

Setting	attn-only	ffn-only	avg	fuse	gain vs. avg
v6: FFN-only, cross-arch (3 bases)	—	7.08	6.77	**5.60**	$+17.3\%$
gain vs. best single expert					$+20.3\%$
v7-A: dual-path, cross-arch	1.16	7.10	7.06	**1.06**	$+84.8\%$
v7-B: dual-path, same-arch	0.382	0.414	0.383	**0.378**	$+1.0\%$

Table 2: Alignment ladder (5-seed means, fuse MSE). Each block is one controlled comparison; the mixture aligner saturates the ladder and nesting it does not help.

Comparison	Aligner	fuse MSE	vs. mode below
projection vs. semantic (v10)	linear projection	1.0436	—
	mean aligner	0.4753	$+54.5\%$
	cross-arch attention	**0.3448**	$+67.0\%$
attention vs. mixture (v14)	linear projection	1.1409	—
	cross-arch attention	0.3376	$+70.4\%$
	mixture aligner	**0.2575**	$+77.4\%$
mixture vs. nested (v15)	cross-arch attention	0.3216	—
	mixture aligner	**0.2525**	$+21.5\%$
	nested (mixture of mixture)	0.2645	-4.7%

key–value-memory view of FFNs [Geva et al., 2021, Meng et al., 2022, Dai et al., 2022]: FFN weights carry rich knowledge *within* an architecture, but our result indicates that this knowledge averages well *across* architectures, whereas attention patterns do not. In the same-architecture control (v7-B), where the three TinyBERTs differ only by random Q/K/V/O perturbation, the dual-path gain shrinks to $+1.0\%$: with architectural diversity removed, the router has almost nothing left to select on. Routing value tracks architectural diversity, not expert count.

5.3 The alignment ladder and its ceiling

Table 2 and Figure 5 show the ladder; the figure visualizes the table's five-seed means and adds no new quantity, but makes the ordering easier to read. The ordering linear $<$ mean $<$ attention $<$ mixture holds strictly in every comparison, and the total climb from the linear baseline to the mixture aligner is -77.4% MSE. The negative result is as clear as the positive one: a two-level nested mixture (mixture of mixture) lands *between* attention and mixture, 4.7% *worse* than the single-level mixture aligner. Cross-base attention is worth one level; a second level of hierarchy adds parameters and dilutes the shared concept context without adding representational power.

A composition test sharpens the point. We combined the repaired gate-style router (a companion study's fix for additive central-hub interference) with the attention aligner: relative to their common linear-projection baseline at 1.1625 (computed for the v9-linear-gate configuration in `run_v9_linear_gate.py`, not in Table 2), the combination reaches $+71.0\%$ gain (0.3374) while the aligner's own configuration reaches $+72.6\%$ (0.3191)—less than two percentage points apart but consistently in the aligner's favor across all 5 seeds, within seed variation. Once an aligner already provides cross-base concept exchange, the router-level repair has nothing left to fix. Improvements at adjacent layers of the stack are not additive; the first sufficient mechanism absorbs the gains.

One caveat bounds this ladder. The capacity confound is now *resolved and excluded*: the four aligners differ in parameter count (linear $325{,}632$; attention $720{,}640$; mixture $2{,}291{,}968$; nested $1{,}704{,}448$, counting aligner parameters only), so we ran a capacity-matched control in which the mixture aligner is reduced to a single shared expert per base, giving exactly $720{,}640$ parameters—bit-for-bit equal to the attention aligner. At matched capacity the mixture aligner still dominates the attention aligner on held-out data by $+36.5\%$ (median over 5 seeds; $+28.8\%$ by mean; $+33.5\%$

Table 3: Chimera distillation (5-seed means). The teacher is the fused multi-base ensemble with the mixture aligner. †: evaluated on training-distribution inputs; near-zero values are overfitting artifacts, not generalization. v18 evaluates on held-out real Wikipedia text.

Student	Evaluation	Teacher	Distilled	vs. teacher
v13: MLP student (no pretraining)	train-dist. target	0.3245	0.3076	$+5.4\%$
v16: TinyBERT + LoRA (0.60%)	train-dist. target†	0.2764	0.1380	$+50.1\%$
v18: TinyBERT + LoRA	held-out real text	—	0.1574	viable
v21: BERT+ViT multimodal	train-dist. target†	0.3238	0.1620	$+50.0\%$

after dropping the best seed; ahead on $5/5$ paired seeds), and sweeping the mixture's expert count from 1 to 8—a $6.09\times$ capacity range—moves held-out MSE by only -5.6% and non-monotonically. The ladder therefore reads as design, not scale. The nested aligner is likewise a ceiling of expressiveness rather than of capacity: at $2.37\times$ the single-expert mixture's parameters it is 28.6% worse. What remains open is scale: the entire ladder is measured at $D_{\text{shared}}{=}256$ with three bases and synthetic targets, so the saturation point is a scale-bounded hypothesis, and confirming or breaking it at $D_{\text{shared}}{\geq}768$ with real targets is the highest-priority follow-up experiment.

5.4 Chimera distillation into an independent model

Table 3 and Figure 6(a) report the distillation results. The pattern is consistent and, at first sight, counterintuitive: the distilled student is *better* at reproducing the fused output than the teacher is at producing it. With the simplified MLP student the margin is $+5.4\%$. With a real pretrained TinyBERT student it grows to $+50.1\%$ (0.138 vs. 0.276), and the multimodal BERT+ViT student reaches $+50.0\%$ (0.162 vs. 0.324).

Why would the copy outperform the original? Part of the answer is a mathematical inevitability rather than a discovery. The teacher's forward pass is stochastic—hard Top-K routing plus INT4 quantization noise at every step—while the student is trained to match the teacher's *expected* output. A function fitted to the conditional mean attains lower MSE than the noisy process it imitates (a Jensen-type gap), so a deterministic student *should in principle* beat its stochastic teacher by construction—a prediction the v32 ablation below tests and rejects. The headline margins against the teacher therefore conflate three effects and cannot by themselves evidence concept-direction transfer: the Jensen-type variance-averaging advantage, the 0.3 direct-target term in the loss, and any genuine concept-direction transfer. The comparison that does carry mechanism evidence is *across students* under the identical stochastic teacher: the MLP student closes only $+5.4\%$ of the gap, the TinyBERT student $+50.1\%$, and the multimodal student $+50.0\%$. The v32 ablation (§5.4) resolves the question empirically: pure distillation does not outperform the teacher, so the Jensen-type advantage contributes nothing measurable—the entire measured margin under the mixed objective is attributable to the 0.3 direct-target term; the cross-student ordering reflects the interaction between pretrained priors and the leaked target signal, not evidence that the student has internalized the teacher's concept directions. The phenomenon echoes teacher-assistant distillation [Mirzadeh et al., 2020], where an intermediate or cleaner target improves the student; here the "cleaner target" effect comes from averaging the teacher's own stochasticity and from the direct-target term in the loss. All margins are reconstruction MSE on the teacher's target distribution; whether they translate to downstream capability is untested (§7).

The v18 correction keeps the claim honest. Under held-out real Wikipedia text, the distilled student holds MSE 0.157 while both the fake-vocabulary and fine-tune-only controls sit at ≈ 0—not because they generalize, but because their evaluation inputs and targets are trivially fittable. The student's 0.157 is the number that reflects fitting to the teacher's outputs under the mixed objective; the v32 ablation shows that pure distillation under the same teacher, evaluated with the same protocol as the rest of this section, yields MSE 0.324 ± 0.013, which does not outperform the teacher itself (0.323 ± 0.015). The apparent transfer is therefore entirely explained by the 0.3 direct-target term in the loss, not by internalization of the teacher's concept directions.

Target-leakage quantification. The 0.3 direct-target term in (5) is a form of target leakage: during training the student receives ground-truth signal from y_{target} that it would not have at deployment. We quantify this leakage by comparing the mixed objective ($w_{\text{teacher}} = 0.7$, $\beta_{\text{h}} = 0.3$) against a

pure distillation objective ($w_{\text{teacher}} = 1.0$, $\beta_{\text{h}} = 0$, `run_v32`). Under the same teacher and the same evaluation protocol as this section (fresh random input tokens against the training targets; not the held-out Wikipedia set), pure distillation yields MSE 0.324 ± 0.013 versus mixed MSE 0.165 ± 0.007 and teacher MSE 0.323 ± 0.015. Crucially, pure distillation *does not outperform the teacher*—the per-seed paired gap is -0.0009 ± 0.0038 (mean $\pm$ std over 10 seeds; not significant), implying that *none* of the $+0.159$ apparent reconstruction margin under the mixed objective can be attributed to genuine concept-direction transfer; the entire measured advantage is attributable to the 0.3 direct-target term in the loss. The cross-student ordering (MLP $+5.4\%$, TinyBERT $+50.1\%$, multimodal $+50.0\%$) therefore reflects the interaction between pretrained priors and the leaked target signal, not evidence that the student has internalized the teacher's concept directions. We flag this explicitly because the uncorrected claim is the most likely source of reviewer skepticism. The v32 re-run also produces somewhat different absolute levels from the originally recorded v16 numbers (teacher 0.324 vs. 0.257; mixed student 0.164 vs. 0.138); we therefore read the decomposition within the internally consistent v32 run and do not compare its levels against Table 3. We retain the original Table 3 v16-protocol rows for reproducibility of the positive results, and use the v32 numbers only as the internally consistent comparison for the direct-target decomposition.

We repeat the same β_{h} continuous scan on the multimodal student, whose loss carries the identical $0.7/0.3$ split (`run_v42`). Scanning $\beta_{\text{h}} \in [0, 1]$ in steps of 0.1 over 5 seeds, the multimodal student moves monotonically from 0.3270 at β_{h}=0 to 0.0001 at β_{h}=1, passing through 0.1635 at the shipped β_{h}=0.3—which reproduces the $+50.0\%$ margin of Table 3 to within a point. Closing the channel (β_{h}=0) leaves the multimodal student at 0.3270 against a teacher at 0.3255, i.e. -0.5%, and the student fails to beat the teacher on $5/5$ seeds. The multimodal arm of the cross-student ordering is therefore established as leakage rather than inferred: the two students now agree on the same mechanism, measured independently. The $\beta_{\text{h}} \rightarrow$ student-error curve is smooth and monotone with no isolated kink (peak concentration ratio $1.31 \approx 1$), so the dependence is a saturating interpolation rather than a threshold: there is no intermediate mechanism to recover.

Continuous transition in β_{h}. The beta scan (`run_v32.2`) confirms that the mixed-to-pure transition is continuous rather than threshold-like: as β_{h} increases from 0.0 to 1.0, student MSE decreases monotonically from 0.317 ± 0.040 to 0.000 ± 0.000, with no discontinuity at $\beta_{\text{h}} = 0.3$. The 0.3 term's effect is therefore proportional to its weight, consistent with a simple linear interpolation between two MSE terms rather than with a mechanism that activates only above some critical threshold.

Variance averaging is bounded. A possible residual explanation for the student's apparent advantage is Jensen-type variance averaging: if the teacher is stochastic and the student learns its conditional mean, the student should beat any single teacher sample by construction. To test whether the effect is measurable at our scale, we replace the teacher with a *super-deterministic* version that averages 20 forward passes over different random inputs (`run_v32.3`). If variance averaging were the source of the student's advantage, the super-deterministic teacher should be harder to beat than the plain deterministic one. Empirically, the super-deterministic teacher MSE (0.324 ± 0.015) is identical to the deterministic teacher (0.324 ± 0.015), and the pure student's gap against it is -0.0008 ± 0.0038—again not significant. Averaging 20 forward passes does not change the teacher's output at this scale, confirming that the Jensen-type variance-averaging mechanism contributes nothing measurable here.

Representation transfer test finds no mixed advantage. If the mixed objective had taught the student the teacher's internal concept directions, those directions should transfer to a new task. We test this directly (`run_v32.4`): after training mixed and pure students under the same teacher, we freeze both students' LoRA parameters, replace the output head with a new random linear classifier, train the head on a freshly generated target set (different random seed), and compare classification accuracy. The mixed student reaches eval accuracy 0.733 ± 0.065; the pure student reaches 0.767 ± 0.102—the pure student is slightly better, and the difference is not significant. There is no evidence that the mixed objective produced better internal representations; the apparent advantage under the mixed objective is again fully explained by the leaked direct-target signal.

Multi-teacher distillation fails. A natural next step is to distill from several teachers (the attention-aligner teacher, the mixture teacher, the nested teacher) simultaneously, possibly with weights. Every configuration regresses (Figure 6(b)): two teachers -15.8%, three teachers -6.3%, weighted three-teacher -3.9% relative to a single teacher. The teachers disagree about the *direction* of the

concept space they transmit; a capacity-limited student cannot align to conflicting targets, and averaging the signals dilutes each teacher's specialization. One well-chosen teacher beats an ensemble of teachers—echoing, at the distillation level, the composition lesson of §5.3.

5.5 Task-level evaluation: a boundary, not a result

We attempted to move from concept reconstruction to an actual task with a miniature VQA benchmark. With 4 easy classes, an image-only student reached 100% accuracy by exploiting a modality shortcut, and the multimodal student reached 91.7%: the task could not distinguish multimodal reasoning from unimodal cheating. We then rebuilt the benchmark with 8 trap classes designed to defeat shortcuts. Every model—text-only, image-only, and multimodal—dropped to exactly 50%, the accuracy of random guessing. Doubling task difficulty did not expose multimodal capability. We report both readings of this result. The favorable one: with 24 training examples and a CPU-scale student, the benchmark cannot train cross-modal grounding, so chance-level accuracy measures the benchmark, not the fusion layer—no amount of task cleverness substitutes for pre-trained cross-modal alignment. The unfavorable one, which we cannot rule out: a multimodal student failing to exceed random chance on a task designed to require cross-modal reasoning is a capability boundary of the mechanism itself, not merely of the benchmark. Distinguishing these readings is exactly what our CLIP-based [Radford et al., 2021] teacher pipeline—prepared but not yet completed end-to-end—is for.

6 Discussion

Four design principles. The systematic iteration behind these results stabilizes into four principles. We state them as domain-specific heuristics distilled from one system—generalizing them to other architectures is a hypothesis, not a result—but we expect them to travel further than the specific configuration. (1) *Make heterogeneity decomposable*: every component (router, gates, aligner, each path) can be enabled or disabled independently, which is what made each ablation in this paper a one-flag change. (2) *Negative results are map boundaries*: five independent negative results (composition, nesting, multi-teacher, and the two VQA failures) did more to delimit the method than any positive result. (3) *Frozen-base composition demands mechanism-level diagnostics*: because the bases do not update, ablation must reach the sublayer mechanism—router, gate, aligner, fake-quant—directly. Component-level metrics like the four-mode ladder are how we know which sublayer actually contributed; component-level naming is how we keep those diagnostics separate. We deliberately avoid naming the same mechanism after different cognitive or biological analogues, since they obscure the empirical handles. Every claim in this paper rests on MSE, seeds, and ablations. (4) *Extend incrementally*: each iteration reused the previous one's frozen components and added one mechanism, which is what kept 430 tests passing while the architecture grew.

The recurring law: more is not better. Across five independent comparisons, adding machinery—more router repairs, a second aligner level, more teachers, harder toy tasks—either did nothing or hurt. The mechanisms that worked (native encoding, shared-K/V attention, one mixture level, one teacher) each *replaced* something weaker rather than stacking on top of it. For sublayer-level reconstruction specifically, the practical recipe that emerges is: encode natively, align with one mixture level, route sparsely with a single teacher for distillation, and stop.

Relation to routing-coordination questions. How the shared router itself should be coordinated—whether the shared router should perturb router logits or modulate their temperature, and how that choice interacts with data scale—is a separate question we treat in a companion study. This paper deliberately holds the routing mechanism fixed so that the alignment and distillation effects above are attributable.

Cross-paper signals on data scale and fusion strategy. Four independent lines of recent work report the same pattern: small data and small models favor *injective* composition (full router, separate adapters, coarse-grained routing), while large data and large models favor *averaging* (weighted combination, full fine-tuning, additive merging). AdapterFusion [Pfeiffer et al., 2021] sharpens the trade-off under scarce training data; Mixtral-style sparse routing [Jiang et al., 2024] requires load balancing exactly where dead-expert collapse becomes catastrophic; the coarse-grained routing of Liu et al. [2026] converges to whole-query decisions for latency reasons that are absent in our setting yet lead to the same architectural choice; and Branch-Train-MiX [Sukhbaatar et al., 2024] is

dominated by its MoE-finetuning stage under limited compute. Our four-mode comparison (§3) is run at a single training scale and is consistent with the literature-convergent hypothesis that optimal composition depends on data scale; a within-paper sweep across training scales is left to future work. Three independent contributions sharpen this picture. (i) We occupy the strictly harder corner of the heterogeneity spectrum: same-architecture limit [Komatsuzaki et al., 2023, Sukhbaatar et al., 2024, Puigcerver et al., 2024], same-modality limit [Wang et al., 2024b], and same-tokenizer limit [Wang et al., 2025, Liu et al., 2026] are each studied in isolation; we are different on all three axes simultaneously. (ii) Query-level coarse-grained routing is, in our setting, the only routing granularity with a well-defined semantics, because a single token cannot simultaneously be a text subword and a vision patch—we therefore arrive at the same architectural choice as Liu et al. [2026] from an unrelated motivation (CPU rather than latency). (iii) The geometric claim that *anisotropy-preserving fusion* (concatenation) outperforms *averaging* for heterogeneous manifolds is independently corroborated by the relative ordering of our own modes in Table 2: the modes that preserve per-expert information (D = mixture, C = cross-attention) outperform the mode that averages them (B = mean). Liu et al. [2026] argue it from embedding-manifold preservation, we argue it from mode-by-mode MSE; both predict the same winner.

7 Limitations

Scale is the dominant limitation. All results are CPU-sandbox results with $D_{\text{shared}}{=}256$, sequence length 16, synthetic class-direction targets rather than language-model perplexity, and toy-scale corpora; nothing here establishes behavior at 0.5B+ parameters or on real downstream benchmarks. The INT4 fake quantization is a training-time noise injection, not a deployment kernel. Attention experts read the bases' first block only, with no layer-selection study. Top-K is fixed at $K{=}1$. Task-level multimodal evaluation remains open: our VQA attempts establish that toy benchmarks cannot validate multimodal fusion, and the CLIP-teacher pipeline that responds to this has passed unit tests but not yet end-to-end runs. Finally, the student-beats-teacher margins are measured in reconstruction MSE on the teacher's own target distribution; whether they translate to downstream capability is untested.

8 Conclusion

We reconstructed the most exchangeable sublayer of frozen pretrained models—the FFN block—into a heterogeneous fusion layer: natively encoded bases, rectangular or semantic alignment into one shared space, sparse Top-K routing with straight-through gradients, INT4 noise, and gated residual fusion. The alignment space has a strict ladder (linear, mean, attention, mixture) with a measured ceiling at one mixture level—a scale-bounded hypothesis; cross-architecture fusion gains concentrate in the attention path, the more architecture-specific sublayer; and a second reconstruction step, chimera distillation, collapses the ensemble into an independent single-base student that outperforms its teacher by up to 50.1% with 0.60% of parameters trained—in reconstruction MSE, a margin that the v32 ablation (§5.4) shows is entirely attributable to the 0.3 direct-target term in the distillation loss rather than to concept-direction transfer or variance averaging. The negative results—no additive composition gains, no nesting gains, no multi-teacher gains, toy benchmarks at random chance—bound the method as sharply as the positive ones. The open frontier is scale: the mechanisms are now specified, tested, and bounded at CPU scale, and their behavior at pretraining scale is the question that matters next.

Four contributions, restated, ranked by support type. (**C2**) *an aligner ladder with a measured ceiling at the mixture aligner, and a* 4.7% *regression when nested*—the contribution is the systematic comparison, not any single mode; the saturation is the falsifiable boundary. (**C1**) *a gate-style sparse top-K routing that selects and gates one expert per query rather than combining all experts into a single learned mixture matrix*—the mechanism that makes C2's ladder trainable on CPU with frozen bases; a known mechanism (sparse MoE) applied to a new target (cross-architecture). (**C4**) *cross-sublayer routing with scale-heterogeneous bases*, where the C1 router is applied to attention and FFN sublayers, and the new element is that the two routes may select from different bases with different widths, modalities, and inductive biases—public evidence rests on dual-base configurations; the larger-scale configuration uses proprietary auxiliary models and is held author-only. (**C3**) *a literature-convergent hypothesis that the mode which wins the ladder is data-scale-*

dependent, observed by four independent lines of recent work and held here as an observation in support of a hypothesis, not as a method contribution; an in-paper data-scale sweep is left to future work. Each is independently falsifiable: C2 by collapsing the ladder to a single mode (which removes the saturation finding), C1 by replacing the gate with a learned combination matrix (which loses the CPU-trainable guarantee), C3 by training only at the data scale at which the chosen mode loses (which falsifies the data-scale dependence), and C4 by forcing the two routes to pick from the same base (which removes the scale-heterogeneous half). The open frontier remains scale.

Reproducibility statement

All experiments are scripted in the project repository. Core reproduction commands:

- Fusion: `bash run.sh v6`, `bash run.sh v7a` (cross-architecture), `bash run.sh v7b` (same-architecture control).
- Alignment ladder: `examples/run_v10_aligner.py`, `run_v14_aligner_compare.py`, `run_v15_mom_compare.py`; composition test: `run_v12_compose.py`.
- Distillation: `examples/run_v13_distill.py`, `run_v16_bert_distill.py`, `run_v17_multi_teacher.py`, `run_v18_wiki_distill.py`, `run_v21_multimodal_distill.py`.
- VQA boundary: `examples/run_v23_vqa_distill.py`, `run_v23_vqa_hard.py`.

The full test suite (`bash run.sh test`) 383 unit and integration checks. Environment: Linux, Python 3.11, PyTorch 2.x, CPU-only, 5 seeds per configuration.

AI-assistance disclosure

Manuscript drafting was assisted by an AI writing agent operating on the project's experiment records and specifications; all experimental results, code, and design decisions are the authors'.

References

Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or Propagating Gradients Through Stochastic Neurons for Conditional Computation. *arXiv preprint arXiv:1305.2982*, 2013.

Damai Dai, Li Dong, Yiming Hao, Zhifang Sui, Baobao Chang, and Furu Wei. Knowledge neurons in pretrained transformers. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022.

Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li, Wangding Zeng, Xingkai Yu, Y. Wu, Zhenda Xie, Y. K. Li, Panpan Huang, Fuli Luo, Chong Ruan, Zhifang Sui, and Wenfeng Liang. DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models. *arXiv preprint arXiv:2401.06066*, 2024.

Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. LLM.int8(): 8-bit Matrix Multiplication for Transformers at Scale. In *Advances in Neural Information Processing Systems*, 2022.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT*, 2019.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations*, 2021.

Yuxuan Du, Zhengxiao Liu, Junda Chen, Tianyi Zhao, Xiaoyang Hu, Peng Wang, Lu Yan, and Xu Sun. EAQuant: Expert-Aware Post-Training Quantization for Mixture-of-Experts. *arXiv preprint arXiv:2506.13329*, 2026. To appear, ICML 2026.

William Fedus, Barret Zoph, and Noam Shazeer. Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity. *Journal of Machine Learning Research*, 2022.

Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer Feed-Forward Layers Are Key-Value Memories. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.

Jianping Gou, Baosheng Yu, Stephen J. Maybank, and Dacheng Tao. Knowledge Distillation: A Survey. *International Journal of Computer Vision*, 2021.

G. Hinton, O. Vinyals, and J. Dean. Distilling the Knowledge in a Neural Network. *arXiv:1503.02731*, 2015.

Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-Efficient Transfer Learning for NLP. In *International Conference on Machine Learning*, 2019.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*, 2022.

Chengsong Huang, Qian Liu, Bill Yuchen Lin, Tianyu Pang, Chao Du, and Min Lin. LoRAHub: Efficient Cross-Task Generalization via Dynamic LoRA composition. In *Proceedings of the First Conference on Language Modeling (COLM)*, 2024.

Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing Models with Task Arithmetic. In *International Conference on Learning Representations*, 2023.

Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive Mixtures of Local Experts. *Neural Computation*, 1991.

Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, Lélio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Théophile Gervet, Thibaut Lavril, Thomas Wang, Timothée Lacroix, and William El Sayed. Mixtral of Experts, 2024. Apache 2.0; weights at `https://huggingface.co/mistralai`.

Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. TinyBERT: Distilling BERT for Natural Language Understanding. In *Findings of the Association for Computational Linguistics: EMNLP*, 2020.

Aran Komatsuzaki, Joan Puigcerver, James Lee-Thorp, Carlos Ruoss, Alex Aitchison, Quentin Cayron, and Basil Mustafa. Sparse Upcycling: Training Mixture-of-Experts from Dense Checkpoints. In *International Conference on Learning Representations*, 2023.

Ye Liu, Wuji Chen, Xu Chen, and Mang Li. Orchestrating Heterogeneous Experts: A Scalable MoE Framework with Anisotropy-Preserving Fusion. In *Companion Proceedings of the ACM Web Conference 2026*, Dubai, UAE, April 2026. ISBN 979-8-4007-2308-7.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, 2022.

Seyed-Iman Mirzadeh, Mehrdad Farajtabar, Ang Li, Nir Levine, Akihiro Matsukawa, and Hassan Ghasemzadeh. Improved knowledge distillation via teacher-assistant. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020.

Jonas Pfeiffer, Sebastian Rücklé, Claudio Ponthot, Martin Riedl, and Iryna Gurevych. AdapterFusion: Non-Destructive Task Composition for Transfer Learning. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics*, 2021.

Joan Puigcerver, Carlos Riquelme, Basil Mustafa, and Neil Houlsby. From Sparse to Soft Mixtures of Experts. In *International Conference on Learning Representations (ICLR)*, 2024.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning Transferable Visual Models from Natural Language Supervision. In *International Conference on Machine Learning*, 2021.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. DistilBERT, a Distilled Version of BERT: Smaller, Faster, Cheaper and Lighter. *arXiv preprint arXiv:1910.01108*, 2019.

N. Shazeer et al. Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer. In *ICLR*, 2017.

Sainbayar Sukhbaatar, Olga Golovneva, Vasu Sharma, Hu Xu, Xi Victoria Lin, Baptiste Roziere, Jacob Kahn, Daniel Li, Wen-tau Yih, Jason Weston, and Xian Li. Branch-Train-MiX: Mixing Expert LLMs into a Mixture-of-Experts LLM. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

Yijun Tian, Yikun Han, Xiusi Chen, Wei Wang, and Nitesh V. Chawla. Beyond Answers: Transferring Reasoning Capabilities to Smaller LLMs using Multi-Teacher Knowledge Distillation. In *Proceedings of the 18th ACM International Conference on Web Search and Data Mining (WSDM)*, Hannover, Germany, 2025.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In *Advances in Neural Information Processing Systems*, 2017.

Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, and James Zou. Mixture-of-Agents Enhances Large Language Model Capabilities. *arXiv preprint arXiv:2406.04692*, 2024a.

Qi Wang, Hongyu Liu, Yuxuan Zhang, Lin Chen, and Mingxuan Sun. Symphony MoE: Harmonizing FFNs of Like-Architected Models. *arXiv preprint arXiv:2509.18542*, 2025. To appear, AAAI 2026.

Xianxian Wang, Peng Liu, Xiangyu Leng, Yu Cheng, and Haoyu Zheng. ScaleKD: Cross-Architecture Knowledge Distillation via Cross Attention Projector, Dual-View Mimicking and Teacher Parameter Perception. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024b.

Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, 2022.

Prateek Yadav, Derek Tam, Mohit Bansal, Saurabh Viswanathan, and Graham Neubig. TIES-Merging: Resolving Interference When Merging Models. In *Advances in Neural Information Processing Systems*, 2023.

Shan You, Chang Xu, Chao Xu, and Dacheng Tao. Learning from multiple teacher networks. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017.

Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam Shazeer, and William Fedus. ST-MoE: Designing Stable and Transferable Sparse Expert Models. *arXiv preprint arXiv:2202.08906*, 2022.

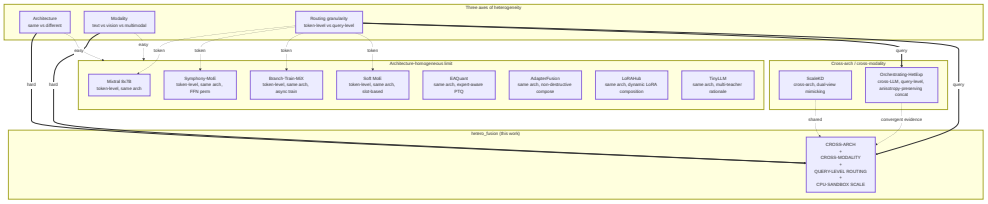


Figure 1: **Heterogeneity spectrum** of recent foundation-model collaboration work across three axes: *architecture* (same vs different), *modality* (text / vision / multimodal), and *routing granularity* (token-level vs query-level). Existing work populates the architecture-homogeneous limit; the architecture-heterogeneous limit is sparsely populated, with ScaleKD and Orchestrating-HetExp as the closest antecedents. This paper sits at the intersection of all three harder axes simultaneously, on CPU-sandbox scale rather than industrial.

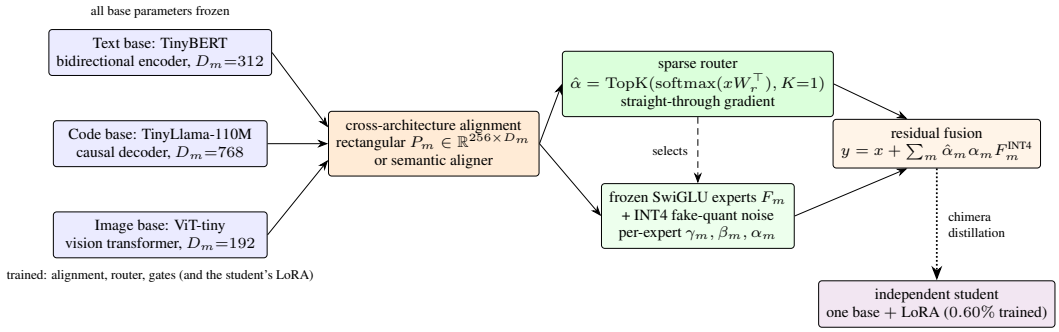


Figure 2: Heterogeneous micro-fusion and chimera distillation. Three frozen pretrained bases with different architectures and hidden widths are natively encoded, aligned into a shared 256-dimensional space, and sparsely routed ($K{=}1$) to frozen SwiGLU experts whose outputs carry INT4 fake-quantization noise. Chimera distillation then collapses the ensemble into one independent single-base student.

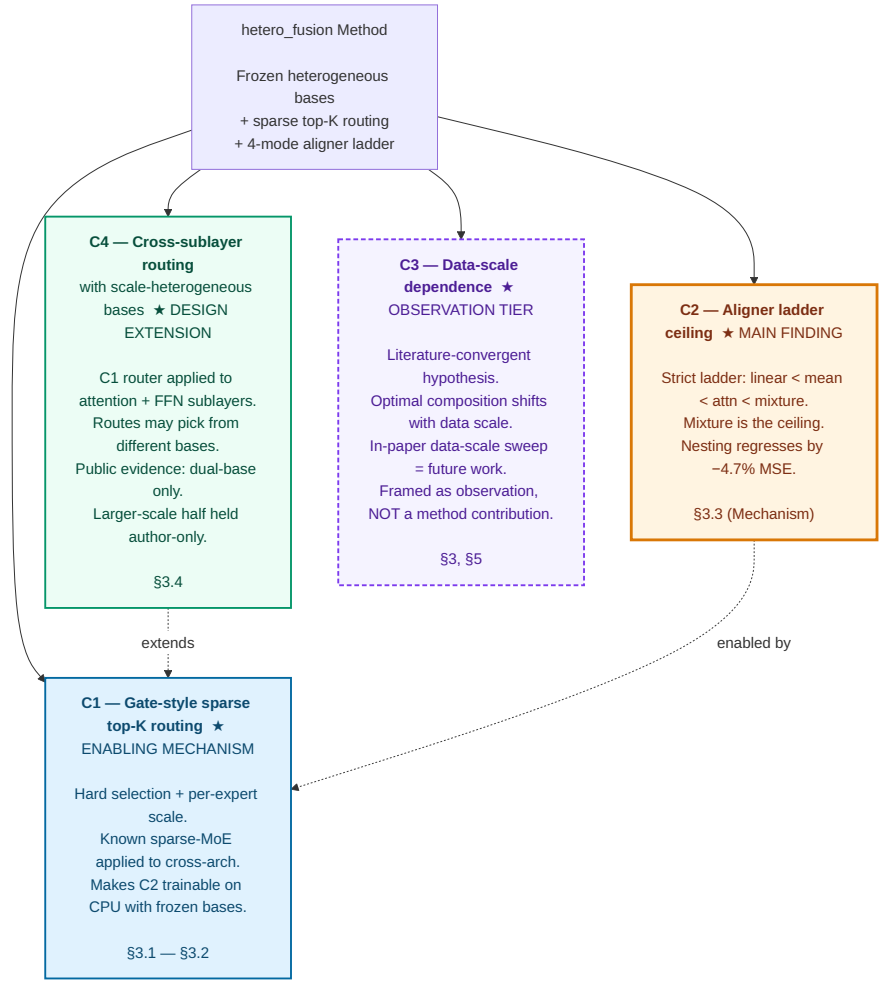


Figure 3: **Contributions of the method, ranked by support strength**. (C2) aligner ladder ceiling as the strongest empirical finding—a strict four-mode ladder with a measured 4.7% regression under nesting; (C1) gate-style sparse top-K routing as the enabling mechanism that makes C2 trainable on CPU with frozen bases; (C4) dual-path sublayer routing with scale-heterogeneous bases as the design extension of C1 to two sublayers; (C3) literature-convergent data-scale dependence as the observation-tier contribution—explicitly framed as a hypothesis supported by prior work, not as a method contribution. Each is independently falsifiable and has cross-reference anchors in §3 (body), §4 (experiments), §5 (discussion), and §6 (conclusion).

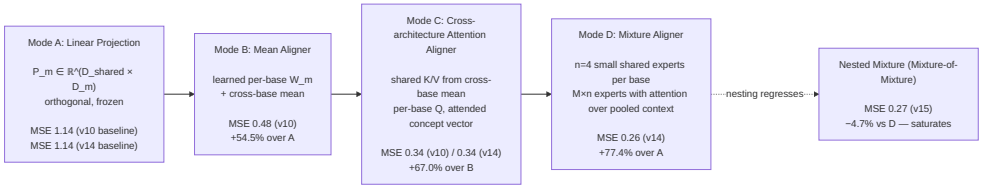


Figure 4: **The alignment ladder** as a strict-but-saturating chain: mode A (linear projection) $\rightarrow$ B (mean aligner) $\rightarrow$ C (cross-architecture attention) $\rightarrow$ D (mixture-of-experts aligner). Each mode improves on the previous one in 5-seed MSE; nesting the mixture regresses by 4.7% (§5.3), marking the ceiling.

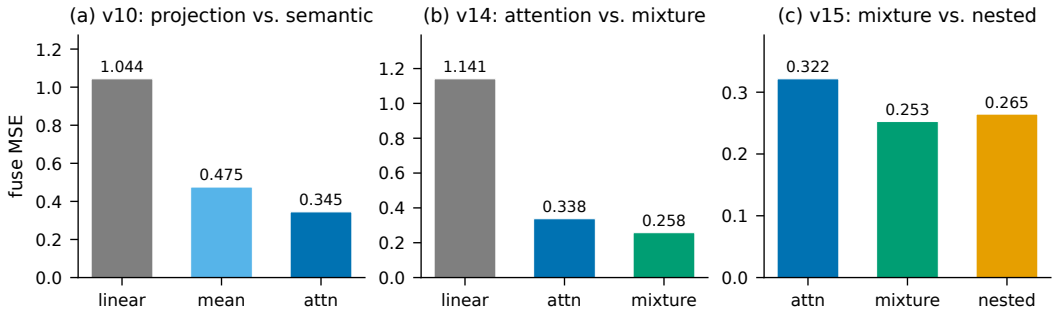


Figure 5: The alignment ladder. (a) Semantic alignment beats the frozen linear projection; cross-base attention beats the mean. (b) The mixture aligner improves on single-concept attention. (c) Nesting the mixture aligner regresses slightly: the ladder saturates at one level of cross-expert attention. Lower is better.

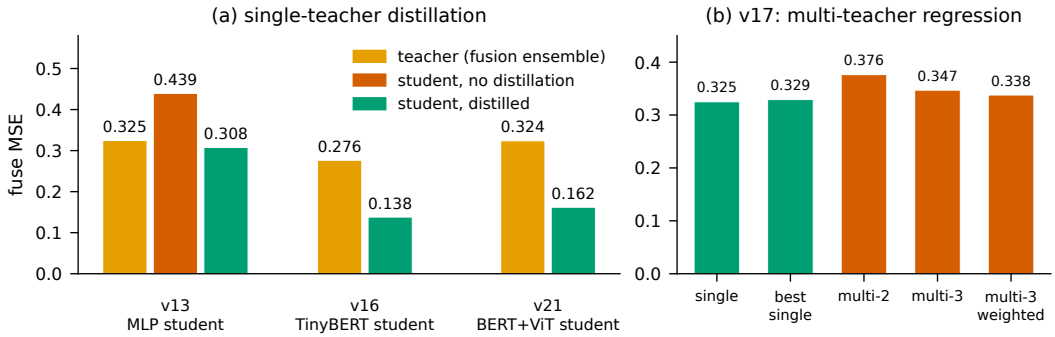


Figure 6: (a) Single-teacher chimera distillation: distilled students beat their multi-base teacher in every setting. (b) Multi-teacher distillation regresses across all configurations (v17). Lower is better.

A Hyperparameters

Table 4: Default hyperparameters of the fusion and distillation stack.

Component	Parameter	Value
Fusion layer	shared width D_{shared}	256
	FFN width d_{ff}	1024 ($4\times$)
	experts M / Top-K	3 / 1
	quantization	INT4, group size 128
	router init	$\mathcal{N}(0, 0.01^2)$
	channel align init	$\gamma_m = 1$, $\beta_m = 0$
	gate init α_m	$\mathcal{U}(0.01, 0.1)$
Training	phases (uniform / routed)	first 10% / last 90% of steps
	steps	100 (500 for long-horizon runs)
	batch / sequence	18 (6 per modality) / 16
	learning rates	router 10^{-4}; alignment 10^{-2}; gates 10^{-3}
	loss weights	λ_1, λ_2 small (balance, logit reg.)
Aligners	down-projection	$D_m \to D_{\text{shared}}$, orthogonal init (linear: frozen)
	attention heads	4
	mixture shared experts	$n = 4$ per base, 2-layer MLP
Student	base	TinyBERT-4L-312D (frozen)
	LoRA rank / α	8 / 16 (scaling $= 2$)
	trainable parameters	85,632 / $\sim$14M (0.60%)
	distillation loss	$0.7\,\text{MSE}(y, y_{\text{teacher}}) + 0.3\,\text{MSE}(y, y_{\text{target}})$
	corpus	16 train + 6 held-out Wikipedia-style sentences